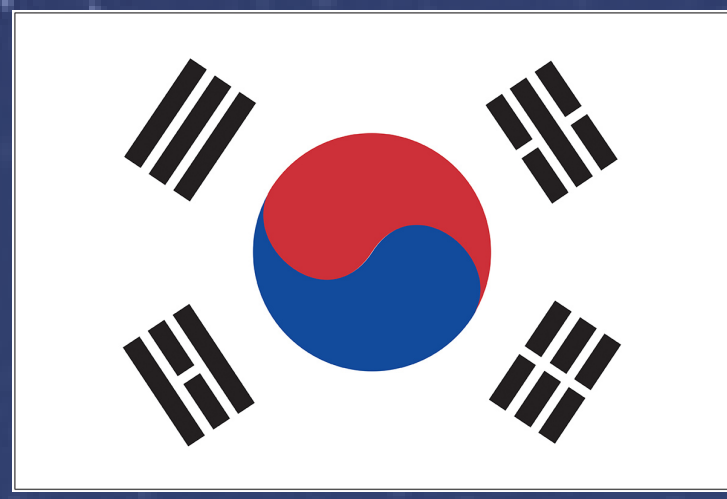




TEAM Smart

RCJ Rescue Line

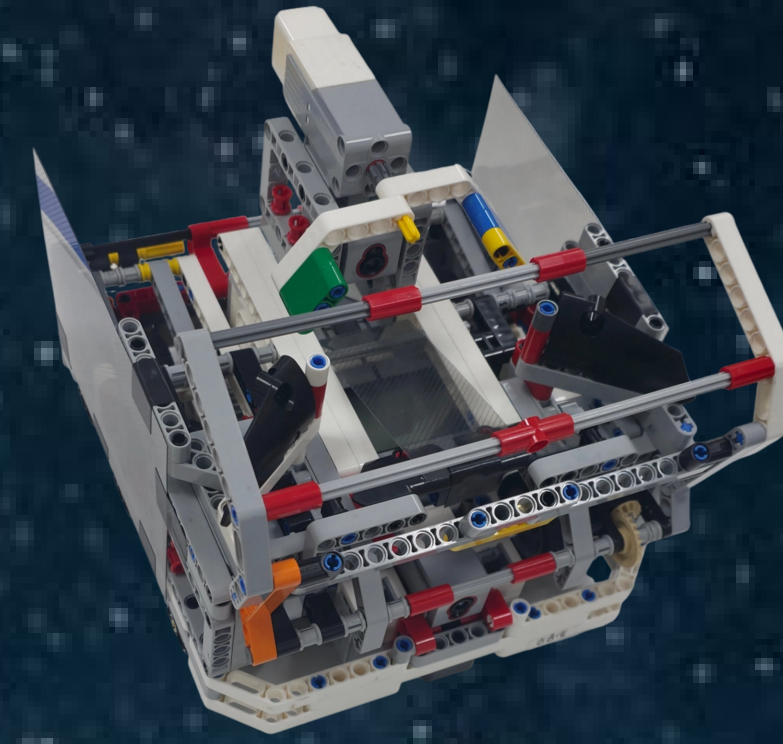


Hardware

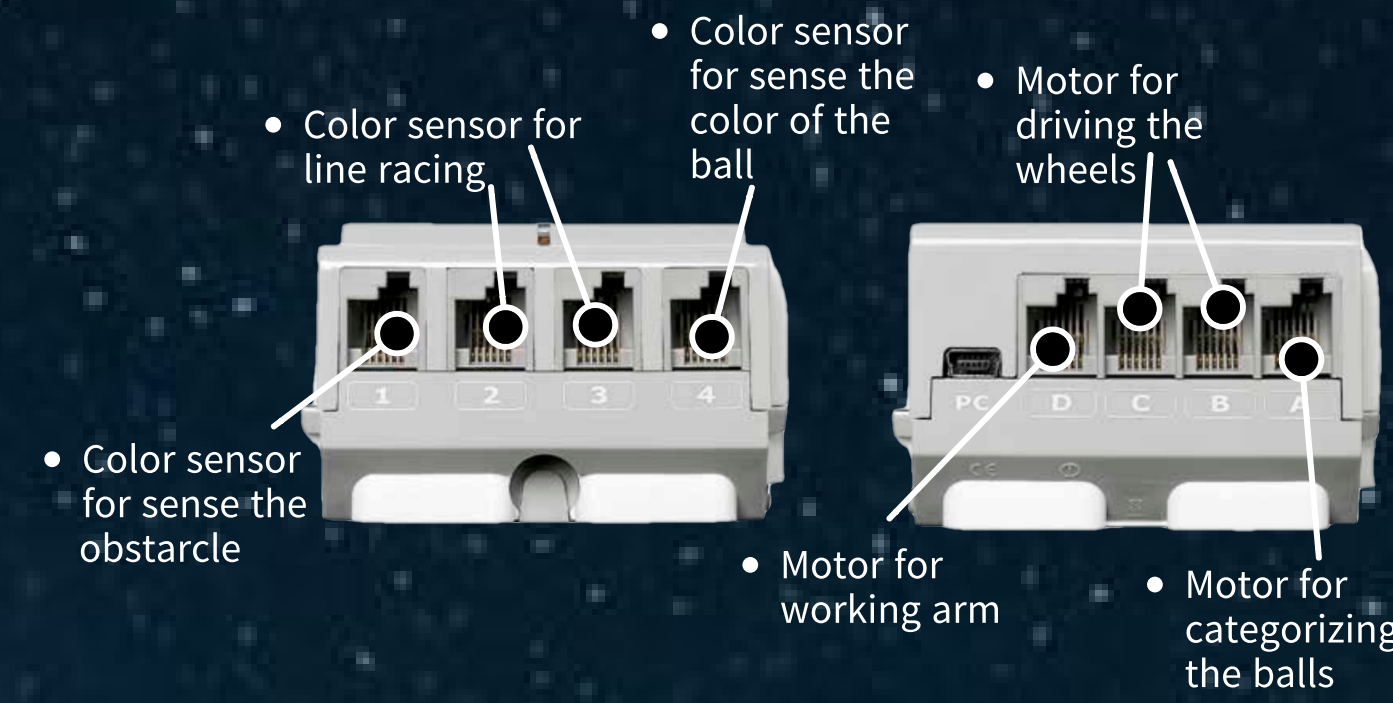
Robot Bottom Assembly



- BETA 0.1 : It was normal overall, but it wasn't smooth on the turn and ramp, so we decided to look for another way
- BETA 0.2 : Used a mehanum wheel for a soft turn, but gave up because it didn't move smoothly on the ramp.
- BETA 0.3 : I used the chain to increase grip, but the grip was so strong that I gave up because the turn motion we expected didn't come out.
- BETA 0.4 : When We tried six-wheels , all the advantages of the Beta 0.1,0.2,0.3 were gathered, so I decided to make six-wheels.
- BETA 0.5 : The use of six wheels in the form of BETA 0.4 often caused situations in which it was not able to cross the bump well. In order to solve this phenomenon, the middle wheel was placed slightly backward to make it easier to cross the bump using the center of gravity.

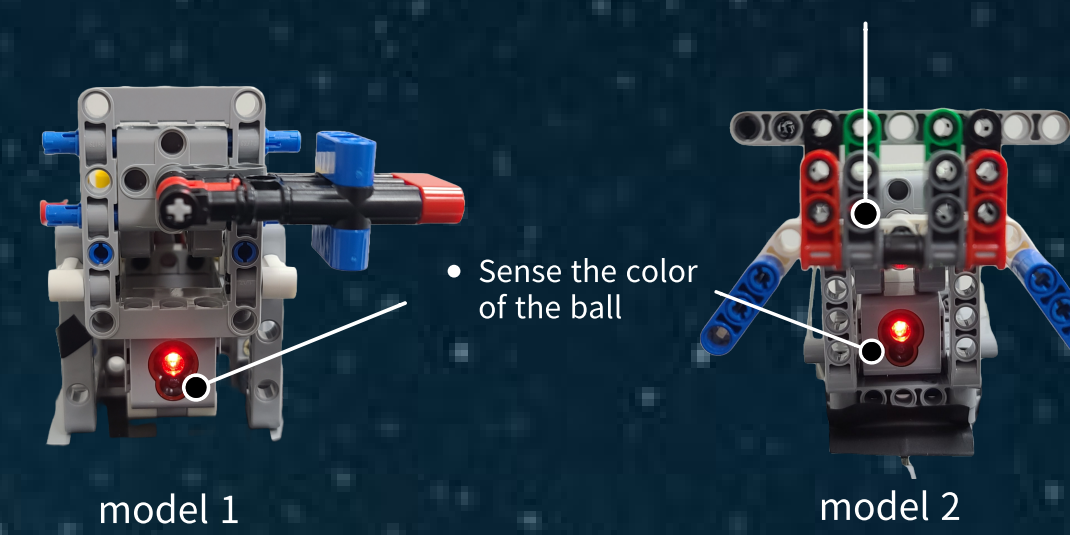


- Six wheels of BETA 0.4 were manufactured using B and C motors to reduce smooth turns and slip on ramps
- Model 2's ball classifier was manufactured with motor A, allowing the position of the ball classifier to be initialized when the program was restarted.
- It was made to do line tracing and evacuation zone detection using sensors 2 and 3.
- After checking the color of the ball using sensor 4, motor A was used to operate the ball sorter to sort the ball.
- The arm was made using a D motor so that the ball could be lifted.



Robot Top Assembly

- Spin the ball sorter 360 degrees in both directions to separate the ball.
- Spin the ball sorter from side to side to separate the ball.



- model 1 : When We restarting the program, We couldn't reset the ball sorter position, so We gave up using it.
- model 2 : When We restarting the program, We could reset the ball sorter position, so We decided using it.

Our Team



Daeyeon

Moojun

Daeyeon Moon : Program, Poster, TDP, Video

I led the overall development of the software, including the evacuation zone and line-tracing programs.

Moojun Kim : Hardware, Poster, TDP, Video

I modified the hardware's ball sorter and wheel alignment to optimize performance.

Green Ratio Test

Green Light(%)	High	Low	Average
1st	65.5	54.5	60
2nd	66.1	55.2	60.6
3rd	64.8	56	60.4
4th	64.3	56.8	60.5

Program Start

Content regarding color sensors 2 and 3 constituting the entire test

Purpose of color sensors 2 and 3 indicating a disputed path marked with a green mark (green measurement)

To verify the content of the area logo with the green marker, please return in the direction of the marker.

Or whether the test has been verified

Verification status: Run the verification program.

Or running the program to identify the type of problem.

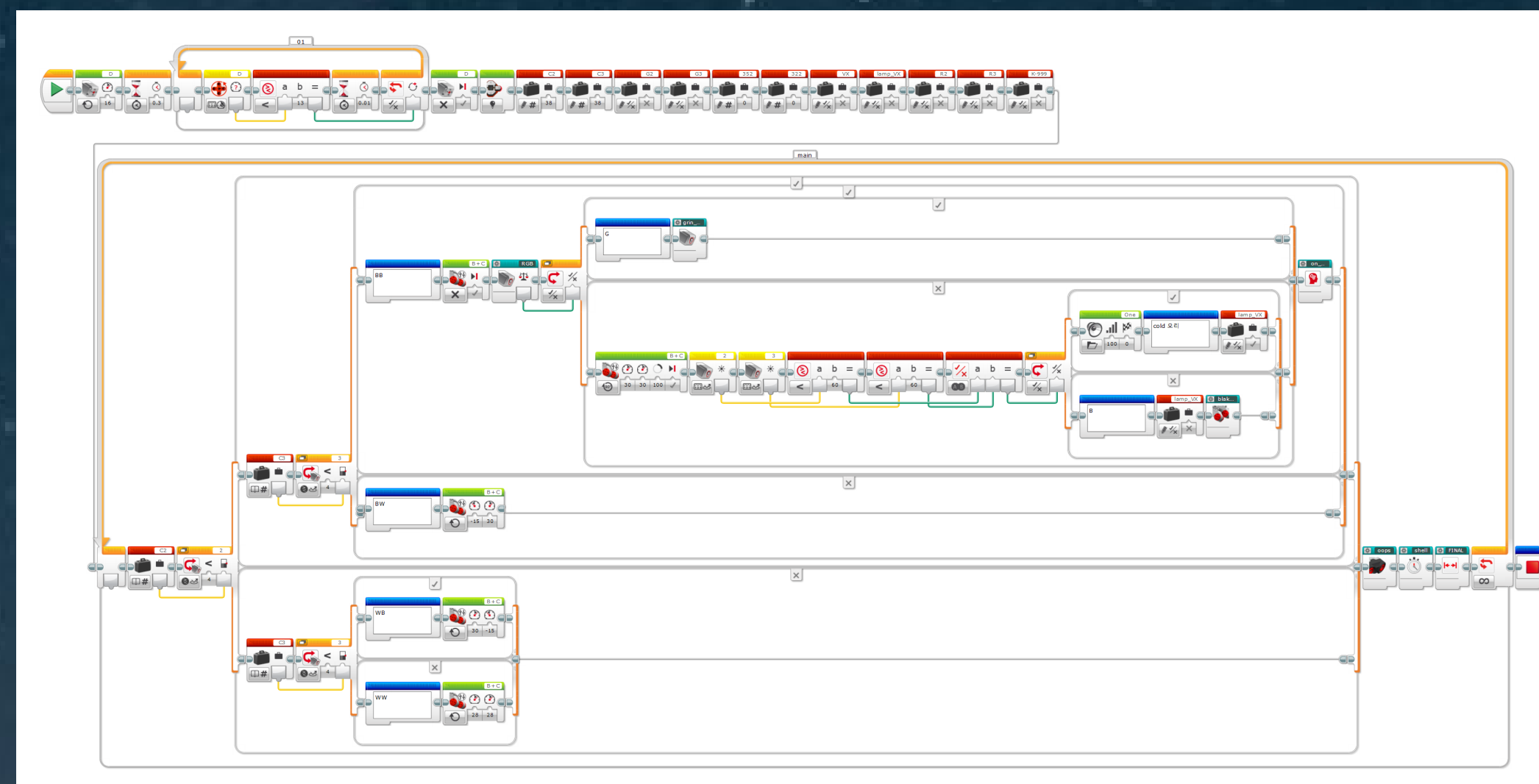
Content regarding the recognition that the distance is separated

Run the training program.

If line tracing found an area contrary to the indication, run the program opposing it, and run the obstacle program that detected an obstacle via sensor 1.

If an end line was detected, terminate the program.

Software



Our Award



2023 Robocup Korea Open Rescue Line Primary

-1st Place

2023 European Championship in Croatia

2024 Robocup Korea Open Rescue Line Secondary

-1st Place

2026 Robocup Korea Open Rescue Line Secondary

-1st Place

Flow - Chart

